

Statistical-inference rigour stack

Each layer filters candidate empirical claims before they enter the headline results.

Layer 0

Per-seed metric values

Per-cell measurements for each dataset, model, method, λ , and metric.



Layer 1

Bootstrap confidence intervals

10,000 resamples; 95% CI for each mean estimate.



Layer 2

Variance-audited effect size

Paired differences with a small variance floor.



Layer 3

Weighting robustness

Equal, voice-heavy, transparency-heavy, consistency-heavy, and rank aggregation.



Layer 4

Family-wise error control

Holm-Bonferroni at $\alpha = 0.05$.

Spearman rank
separability test

Per dataset:
 $H_0: \rho \geq 0.7$
Reject if the
upper CI < 0.7

Checked with
percentile CI,
BCa CI, Holm,
and BH-FDR

Headline empirical claims

Weighting-robust pairwise procedural gaps + 4/4 separability rejection on the primary diagnostic set